MODULE 1 · LECTURE 25

# AI in Crop Health Monitoring and Intelligent Pesticide Application

*How sensors and deep learning detect diseases, pests and weeds; why a model that is excellent in the laboratory can fail in the field; how AI detections are combined with economic threshold levels; and how spot spraying, variable-rate sprayers and drones put pesticide only where it is needed — with worked examples and a spot-sprayer program.*

### Learning objectives

After this lecture you should be able to:

1. Explain crop stresses and the sensors and platforms used to monitor crop health.
2. Describe how deep learning classifies, detects and segments diseases, pests and weeds in images.
3. Explain why AI models trained in the laboratory may fail in the field, and how to make them reliable.
4. Combine AI detections with economic threshold levels to decide when control is needed.
5. Describe spot and patch spraying, variable-rate canopy spraying and drone spraying.
6. Calculate threshold decisions, application rates and herbicide savings with worked examples.

Scope: Lecture 15 explained remote sensing and NDVI, Lecture 12 sprayer calibration and rate control, Lecture 14 the National Pest Surveillance System app, and Lecture 24 machine learning and its evaluation. This lecture applies them to keeping crops healthy and using pesticides only where and when they are needed. Weeding robots follow in Lecture 27.

## 1. Monitoring crop health

Crops suffer **biotic** stress from fungi, bacteria, viruses, insects, mites, nematodes and weeds, and **abiotic** stress from shortage of water or nutrients, heat and salinity. Early and accurate detection and diagnosis are key to reducing yield and quality losses. Optical sensors — RGB imaging, multispectral and hyperspectral sensors, thermography and chlorophyll fluorescence, and more recently 3D scanning — can identify and quantify plant diseases automatically and objectively, at early stages of an epidemic, from single leaves to whole fields (Mahlein, 2016).

| Sensor | What it measures | Typical use |
|---|---|---|
| RGB (colour) camera | Visible symptoms: spots, discolouration, insects, weeds | Smartphone diagnosis, trap cameras, weed detection on sprayers |
| Multispectral camera | Reflectance in a few bands, including near-infrared; indices such as NDVI (Lecture 15) | Drone and satellite maps of crop vigour and stress patches |
| Hyperspectral sensor | Reflectance in hundreds of narrow bands | Detecting subtle changes in pigments and leaf structure; research and phenotyping |
| Thermal camera | Leaf and canopy temperature | Water stress and changes in transpiration caused by infection |
| Chlorophyll fluorescence | Efficiency of photosynthesis | Changes in photosynthesis caused by disease or other stress |
| 3D scanning (LiDAR, stereo cameras) | Plant height, shape and canopy volume | Plant structure; canopy-adapted spraying in orchards |

These sensors are carried on several **platforms**: hand-held devices and smartphones, insect traps, tractors, sprayers and ground robots, drones, and satellites. Closer platforms see finer detail; higher platforms cover larger areas (Lecture 15).

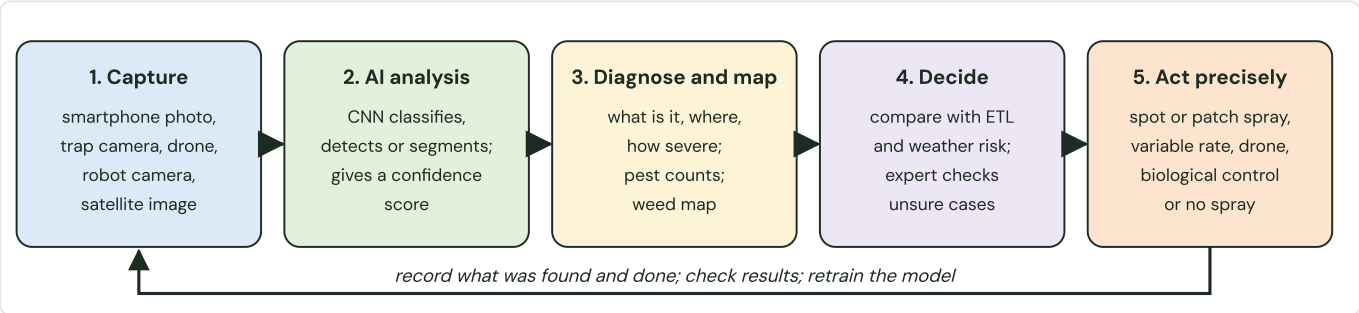



*Figure 1. From image to action: the AI crop-protection pipeline. Original diagram.*

## 2. AI for detecting diseases, pests and weeds

Computer vision with deep learning (Lecture 24) performs three kinds of task on crop images:

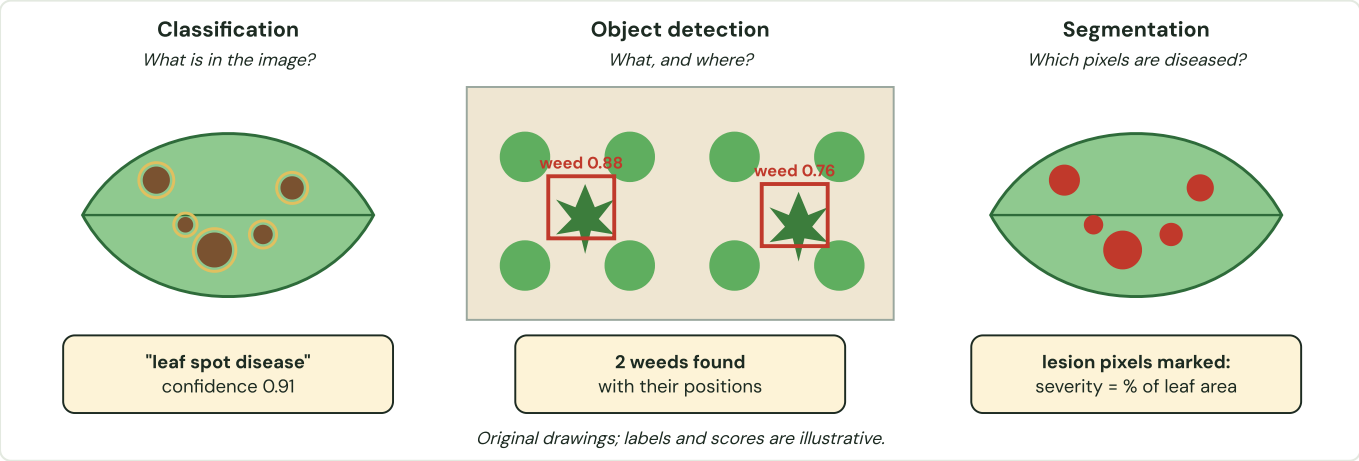
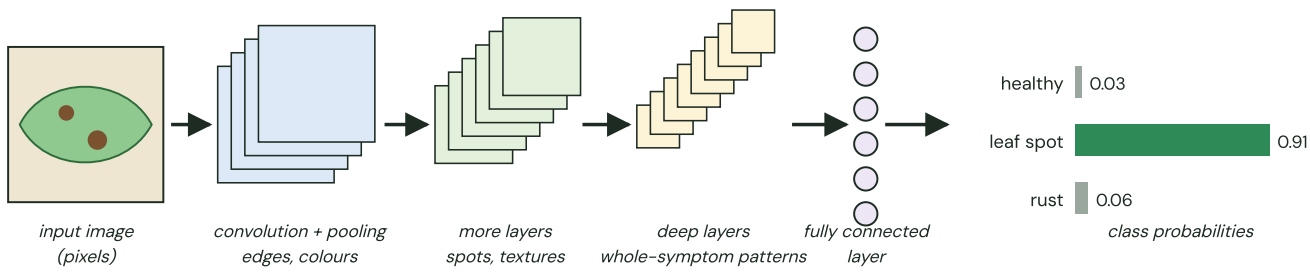



*Figure 2. Three computer-vision tasks used in crop protection. Original drawings.*

| Task | Output | Example |
|---|---|---|
| Image classification | One label for the whole image, with a confidence score | A farmer photographs a leaf; the app names the disease |
| Object detection | A box and a label for each object found | Counting insects on a sticky trap; locating each weed for a spot sprayer |
| Semantic segmentation | A label for every pixel | Measuring the share of leaf area covered by lesions (disease severity) |

For example, if segmentation marks 9,000 of a leaf's 120,000 pixels as lesion, the diseased area is 9,000 ÷ 120,000 = 7.5% of the leaf — an objective severity value that can be tracked over time.

### How a convolutional neural network sees




*Schematic of a convolutional neural network (CNN). Original diagram; layer sizes and output values are illustrative.*

*Figure 3. A convolutional neural network turns pixels into class probabilities. Original schematic.*

A **convolutional neural network (CNN)** passes the image through layers of small filters. Early layers respond to edges and colours, deeper layers to spots, textures and finally whole symptom patterns; the last layer gives a probability for each class. The filters are not written by people: they are learned from thousands of labelled images. A survey of deep learning in agriculture found that it gave high accuracy and outperformed commonly used image-processing techniques (Kamilaris & Prenafeta-Boldú, 2018).

### Laboratory success, field reality

Mohanty et al. (2016) trained a deep CNN on 54,306 images of healthy and diseased leaves of 14 crops with 26 diseases, photographed under controlled conditions. It reached **99.35%** accuracy on held-out test images from the same collection. But on images taken from other trusted sources, under different conditions, accuracy fell to **just above 31%** across the 38 classes — and the model could handle only single leaves, facing up, on a plain background. A model is only as good as the variety of data it was trained on.

- Train on field photographs from many places, seasons, varieties, growth stages and cameras.
- Include healthy plants and look-alike problems — nutrient deficiency, herbicide injury, insect feeding — so they are not mistaken for disease.
- Test on images from new farms that the model has never seen (Lecture 24).
- Show the confidence score and let the app say "not sure — consult an expert"; keep extension experts in the loop.
- Collect corrected cases and retrain the model regularly.

### Counting pests in traps

Pheromone traps attract male moths so that their numbers can be monitored, but counting them by hand is slow. Ding and Taylor (2016) built a deep-learning pipeline that identifies and counts codling moths in trap images, and needs no pest-specific engineering, so it can be adapted to other species. In India, ICAR's Central Institute for Cotton Research (CICR) piloted AI-powered pheromone traps for pink bollworm with 18 farmers in Punjab: a camera in each trap photographs the captured moths, a machine-learning model identifies and counts the pink bollworms, and farmers receive regular updates on their phones (Shagun, 2024). Farmers can also upload photos of pests for expert advice through the National Pest Surveillance System (Athrady, 2024; Lecture 14).

## 3. From detection to decision: thresholds

Finding a pest does not mean that spraying pays. The **integrated control** concept of Stern et al. (1959) uses chemical control only as necessary, in the way least disruptive to natural enemies, and works best when sound economic thresholds, rapid sampling methods and selective insecticides are available. Two levels guide the decision:

| Term | Meaning |
|---|---|
| Economic injury level (EIL) | The lowest pest population at which the expected crop loss equals the cost of controlling the pest |
| Economic threshold level (ETL) | The population at which control should start, so that the pest does not reach the EIL |

AI makes the "rapid sampling" part cheap and continuous: cameras count pests every day, and the decision rule compares the counts with the ETL. For pink bollworm in cotton, ICAR-CICR advises installing pheromone traps at 5 per hectare from 45 days after sowing; the ETL is **8 moths per trap on 3 consecutive days**, or 10% damaged (rosette) flowers, or 10% damaged green bolls — at least 2 of 20 bolls sampled (ICAR-Central Institute for Cotton Research [ICAR-CICR], 2019).

> **Worked example 1 — when does an AI trap reach the ETL?**
> Average moths per trap counted by the trap cameras on days 1–10: 2, 3, 5, 9, 7, 8, 10, 12, 6, 4.
> Day 4 (9 moths) is above 8, but day 5 (7) breaks the run — no action yet.
> Days 6, 7 and 8 give 8, 10 and 12: three consecutive days at or above 8 → the **ETL is reached on day 8**, and an alert advises the farmer to inspect the crop and apply the recommended control.
> A person checking traps once a week could easily miss this window; the camera checks every day.

Weather-based forecasting adds a second line of evidence: many fungal diseases need leaf wetness and humid, mild weather, so IoT weather stations (Lecture 23) can warn when conditions favour infection, and protective sprays can be timed to the risk.

## 4. Intelligent pesticide application

Conventional spraying applies the same dose over the whole field, although weeds, pests and diseases usually occur in patches. Applying the chemical only where it is needed saves money and reduces residues and environmental load.

### Spot and patch spraying

Allmendinger et al. (2022) describe three essential parts shared by all precision spraying systems: **(1) sensors and classifiers** that detect weeds and crop, **(2) decision algorithms** that decide whether control is needed and choose the herbicide and rate, and **(3) a precise sprayer** with boom-section or

single-nozzle control. Simple "green-on-brown" sensors separate green plants from soil and residues; "green-on-green" systems must tell weeds from crop plants, usually with CNNs.

- **Patch spraying** treats weed patches with boom sections; results show at least 50% herbicide saving in various crops without extra weed-control costs in later years (Gerhards et al., 2022).
- Reported savings range from 23% to 89% in cereals, maize, sugar beet and peas, and can reach up to 99% for **spot spraying** of single plants when weeds are few — savings depend on weed density (Allmendinger et al., 2022).
- Cameras can also guide hoes between and within rows, replacing herbicide with mechanical weeding (Gerhards et al., 2022; Lecture 27).

### Variable-rate spraying of tree canopies

In orchards and nurseries a fixed sprayer wastes spray in gaps between trees and on small trees. Chen et al. (2012) built a sprayer whose laser scanner measures the presence, height, width and foliage density of each tree in real time; each nozzle has a **pulse-width-modulated (PWM)** solenoid valve that adjusts its output to the canopy in front of it. Spray coverage inside trees of different sizes stayed similar, showing the potential to reduce spray volume and drift.

A PWM valve opens and shuts many times a second; the fraction of time it is open — the **duty cycle** — sets the average flow, so the rate can be changed nozzle by nozzle without changing pressure.

> **Worked example 2 — application rate with PWM nozzles** (calibration formula from Lecture 12)
> Nozzle flow when fully open 0.8 L/min, nozzle spacing 0.5 m, forward speed 6 km/h.
> Rate = 0.8 × 600 ÷ (6 × 0.5) = **160 L/ha** at 100% duty cycle.
> At 50% duty cycle the average flow is about half, ≈ 0.4 L/min → **≈ 80 L/ha**: the controller can halve the rate over a small tree and switch the nozzle off over a gap.

### Drones for spraying in India

The Ministry of Agriculture and Farmers Welfare released Standard Operating Procedures (SOPs) for using drones to apply pesticides and nutrients in December 2021. They cover statutory provisions, flying permissions, area and distance restrictions, weight classification, drone registration, safety insurance, pilot certification, operation planning, air-flight zones, weather conditions, procedures before, during and after operation, and emergency handling; drones had first been used in India against locust attacks (Press Information Bureau [PIB], 2021). Crop-specific SOPs for applying pesticides with drones followed in April 2023 (PIB, 2023). AI adds value when the drone's spray map comes from a disease or weed map rather than covering the whole field.

## 5. Worked example: an AI spot sprayer

The program below simulates a 20 m × 30 m field divided into 0.5 m × 0.5 m cells, one per nozzle position. Weeds grow in four patches. A trained CNN gives each cell a weed score; weeds usually score high and bare soil or crop low, but the scores overlap, as with any real model. The sprayer opens a nozzle when the score reaches a threshold.

### Simulating an AI spot sprayer in Python (Lecture 8)

```
import random

random.seed(11)
ROWS, COLS = 40, 60                           # field cut into 0.5 m x 0.5 m cells: 20 m x 30 m
PATCHES = [(8, 12), (26, 42), (32, 10), (11, 50)]   # centres of weed patches (row, column)

# 1. Weeds grow in patches: dense in the centre, thinning out, a few scattered elsewhere
weed = []
for r in range(ROWS):
    row = []
    for c in range(COLS):
        d = min(((r - pr) ** 2 + (c - pc) ** 2) ** 0.5 for pr, pc in PATCHES)
        p = 0.7 if d < 4 else 0.25 if d < 8 else 0.02
        row.append(random.random() < p)
    weed.append(row)

# 2. A trained CNN looks at each cell's image and returns a weed score from 0 to 1
def cnn_score(is_weed):
    s = random.gauss(0.72, 0.15) if is_weed else random.gauss(0.28, 0.15)
    return min(max(s, 0.0), 1.0)

score = [[cnn_score(weed[r][c]) for c in range(COLS)] for r in range(ROWS)]

# 3. Open the nozzle over a cell only when its score reaches the threshold
cells = ROWS * COLS
n_weeds = sum(sum(row) for row in weed)
print(f"Field: {cells} cells, weeds in {n_weeds} ({100 * n_weeds / cells:.1f}%)")
print("threshold  area sprayed  herbicide saved  weeds hit  spray on weeds")
for th in (0.3, 0.4, 0.5, 0.6, 0.7):
    sprayed = hit = 0
    for r in range(ROWS):
        for c in range(COLS):
            if score[r][c] >= th:
                sprayed += 1
                hit += weed[r][c]
    area = sprayed / cells
    print(f"   {th:.1f}        {100 * area:5.1f}%         {100 * (1 - area):5.1f}%"
          f"        {100 * hit / n_weeds:5.1f}%      {100 * hit / sprayed:5.1f}%")

Output:
Field: 2400 cells, weeds in 294 (12.2%)
threshold  area sprayed  herbicide saved  weeds hit  spray on weeds
   0.3         52.1%          47.9%        100.0%       23.5%
   0.4         31.2%          68.8%         99.0%       38.9%
   0.5         17.6%          82.4%         92.2%       64.2%
   0.6         11.0%          89.0%         79.6%       88.6%
   0.7          6.5%          93.5%         51.7%       98.1%
```

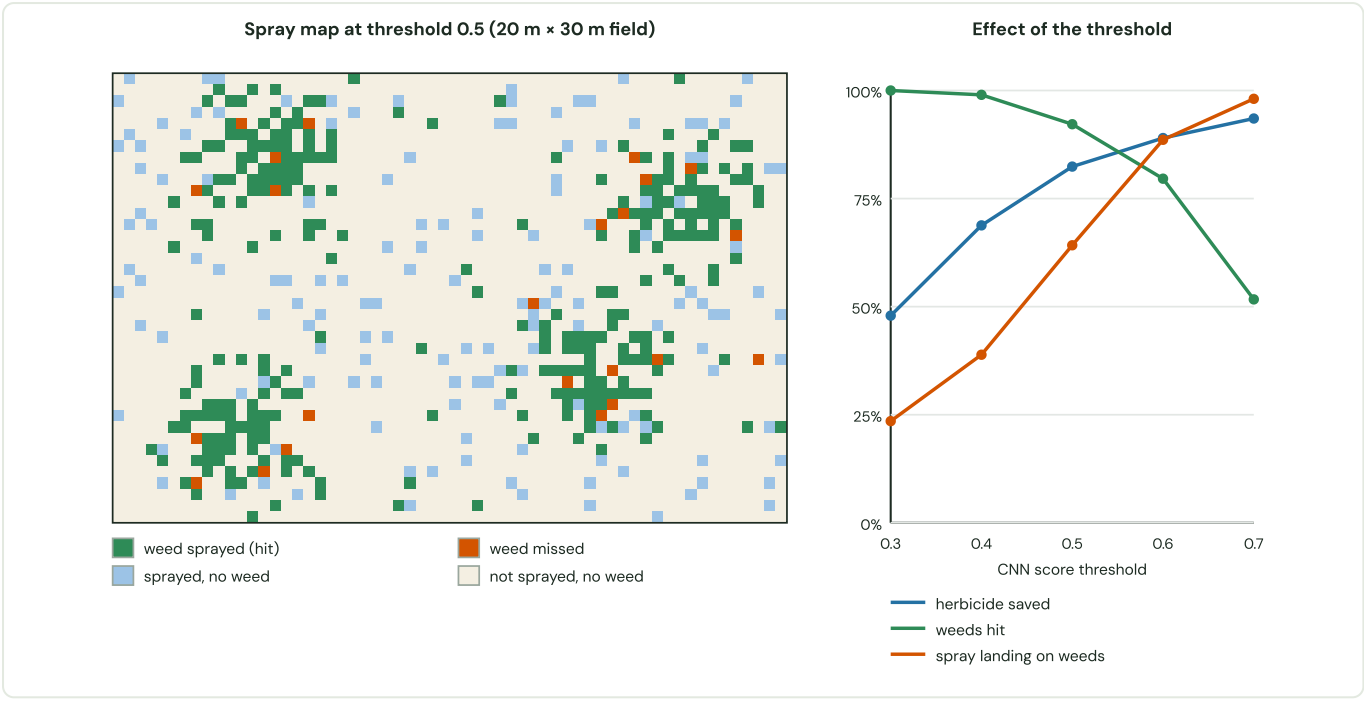



*Figure 4. Left: which cells were sprayed at threshold 0.5. Right: herbicide saved, weeds hit and the share of spray landing on weeds at each threshold. Original chart from the program's data.*

**Worked example 3 — choosing the threshold**

Scale the result up to a 6 ha field with the same weed pattern: blanket spraying at 1.0 L of herbicide product per hectare uses 6.0 L.

At threshold 0.5 the sprayer covers 17.6% of the area → 6.0 × 0.176 ≈ **1.06 L**, a saving of 82.4%, while still hitting 92.2% of weed cells.

At 0.3 nearly all weeds are hit (100%) but half the field is sprayed; at 0.7 only 6.5% is sprayed but half the weeds (48.3%) escape.

Weeds that escape will set seed, so a farmer would choose a threshold near 0.4–0.5 here, or add a buffer around detected weeds.

*The program was run and the output shown is its actual output. The field, weed patches and CNN scores are simulated with illustrative values; the shape of the trade-off — more savings, more escapes — is the same as in real spot spraying, where savings depend on weed density.*

## 6. Benefits and challenges

| Benefits | Challenges |
|---|---|
| Earlier, objective detection of diseases and pests (Mahlein, 2016) | Models trained on laboratory images may fail on field images (Mohanty et al., 2016) |
| Continuous pest counts without daily trap visits (Ding & Taylor, 2016; Shagun, 2024) | Look-alike symptoms: deficiencies, herbicide injury and diseases can resemble each other |
| Sprays timed to thresholds and weather risk rather than the calendar | Need for large, locally labelled image sets for Indian crops, varieties and pests |
| Large herbicide and pesticide savings with spot and patch spraying (Gerhards et al., 2022) | Sensors, controllers and sprayers often use their own protocols and do not connect easily (Gerhards et al., 2022) |
| Less residue, drift and exposure of operators | Cost of cameras, PWM sprayers and drones; regulations and trained pilots for drones (PIB, 2021) |
| Records of every detection and treatment for traceability | Farmers must trust the advice: explanations, confidence scores and expert back-up matter |

## 7. Key takeaways

- Optical sensors from RGB to hyperspectral, thermal and fluorescence detect crop stress, on platforms from phones to satellites.
- Deep learning (CNNs) classifies, detects and segments diseases, pests and weeds from labelled images.
- A model with 99% accuracy in the lab fell to about 31% on field-type images: diverse training data and field testing are essential.
- AI traps count pests continuously; decisions should compare counts with the economic threshold level (e.g. 8 pink bollworm moths per trap for 3 days).
- Spot and patch spraying need detection, a decision algorithm and nozzle-level control; reported herbicide savings are at least 50% for patch spraying and up to 99% for spot spraying.
- PWM nozzles and canopy sensors adjust the spray to each tree; drones in India operate under SOPs issued in 2021 and crop-specific SOPs from 2023.
- The detection threshold trades chemical saved against pests or weeds missed.

## 8. Examination questions

### Short-answer questions (2 marks)

**Q1. Name four optical sensors used to detect plant diseases.**

RGB camera, multispectral/hyperspectral sensor, thermal camera, chlorophyll fluorescence (also 3D scanning).

**Q2. Distinguish image classification and object detection.**

Classification gives one label for the whole image; detection gives a box and label for each object.

**Q3. What does semantic segmentation provide in disease assessment?**

A label for each pixel, so the diseased share of leaf area (severity) can be measured.

**Q4. What is a convolutional neural network?**

A deep neural network whose layers of learned filters extract features from images to classify them.

**Q5. Why did the PlantVillage-trained model perform poorly on other images?**

It was trained on single leaves on plain backgrounds under controlled conditions; accuracy fell to about 31% on images taken under different conditions.

**Q6. Define economic threshold level.**

The pest population at which control should start to prevent it reaching the economic injury level.

**Q7. State the ETL of pink bollworm based on pheromone traps.**

8 moths per trap on 3 consecutive days (ICAR-CICR).

**Q8. Name the three parts of a spot-spraying system.**

Sensors and classifiers; decision algorithm; precise sprayer with section or single-nozzle control.

**Q9. What is a PWM nozzle?**

A nozzle with a solenoid valve pulsed on and off; the duty cycle sets the average flow.

**Q10. Distinguish green-on-brown and green-on-green detection.**

Green-on-brown finds green plants on soil; green-on-green tells weeds from crop plants.

## Long-answer questions (5–10 marks) — answer outlines

**Q1. Describe sensors and platforms used for crop health monitoring.**

Biotic and abiotic stress; sensor table (Mahlein, 2016); platforms; Figure 1 pipeline.

**Q2. Explain how deep learning is used to detect plant diseases, pests and weeds.**

Three tasks (Figure 2); CNN (Figure 3); severity example; Mohanty et al. results; trap counting.

**Q3. Discuss the challenges of using AI disease-diagnosis models in farmers' fields and how to overcome them.**

Lab-to-field gap (99.35% vs ~31%); diverse data; look-alikes; testing on new farms; confidence and experts; retraining.

**Q4. Explain the economic threshold concept and how AI supports threshold-based pest management, with an example.**

Integrated control (Stern et al., 1959); EIL and ETL; pink bollworm ETL; worked example 1; AI traps.

**Q5. Describe intelligent pesticide application technologies.**

Spot and patch spraying parts and savings; canopy sensing and PWM (Chen et al., 2012); worked example 2; drones and SOPs.

**Q6. Explain the trade-off in choosing the detection threshold of an AI spot sprayer.**

Program, output and Figure 4; worked example 3; savings vs escapes; precision and recall (Lecture 24).

## 9. Glossary

| Term | Meaning |
|---|---|
| Abiotic stress | Stress from non-living causes: water, nutrients, heat, salinity |
| Biotic stress | Stress from living organisms: pathogens, insects, weeds |
| CNN | Convolutional neural network: deep learning model for images |
| Duty cycle | Fraction of time a PWM valve is open |
| Economic injury level (EIL) | Pest level at which loss equals the cost of control |
| Economic threshold level (ETL) | Pest level at which control should begin |
| Generalisation | A model's ability to work on data unlike its training data |
| Green-on-green | Detecting weeds among crop plants |
| Hyperspectral sensor | Sensor recording hundreds of narrow wavebands |
| Object detection | Finding and boxing each object in an image |
| Patch spraying | Spraying weed patches with boom sections |
| Pheromone trap | Trap baited with an insect sex attractant for monitoring |
| PWM | Pulse-width modulation: rapid on–off control of a valve |
| Segmentation | Labelling every pixel of an image |
| Spot spraying | Spraying individual plants with single-nozzle control |
| SOP | Standard operating procedure |

## References


Allmendinger, A., Spaeth, M., Saile, M., Peteinatos, G. G., & Gerhards, R. (2022). Precision chemical weed management strategies: A review and a design of a new CNN-based modular spot sprayer. *Agronomy, 12*(7), 1620. https://doi.org/10.3390/agronomy12071620

Athrady, A. (2024, August 16). Agriculture ministry launches national app to collect info on pest attacks in crops. *Deccan Herald*. https://www.deccanherald.com/india/agriculture-ministry-launches-national-app-to-collect-info-on-pest-attacks-in-crops-3153384

Chen, Y., Zhu, H., & Ozkan, H. E. (2012). Development of a variable-rate sprayer with laser scanning sensor to synchronize spray outputs to tree structures. *Transactions of the ASABE, 55*(3), 773–781. https://doi.org/10.13031/2013.41509

Ding, W., & Taylor, G. (2016). Automatic moth detection from trap images for pest management. *Computers and Electronics in Agriculture, 123*, 17–28. https://doi.org/10.1016/j.compag.2016.02.003

Gerhards, R., Andújar Sanchez, D., Hamouz, P., Peteinatos, G. G., Christensen, S., & Fernandez-Quintanilla, C. (2022). Advances in site-specific weed management in agriculture—A review. *Weed Research, 62*(2), 123–133. https://doi.org/10.1111/wre.12526

ICAR-Central Institute for Cotton Research. (2019). *Pink bollworm management strategies for kharif 2019* [Circular]. Indian Council of Agricultural Research. https://icar.org.in/sites/default/files/Circulars/PBW-Management-strategies-2019.pdf

Kamilaris, A., & Prenafeta-Boldú, F. X. (2018). Deep learning in agriculture: A survey. *Computers and Electronics in Agriculture, 147*, 70–90. https://doi.org/10.1016/j.compag.2018.02.016

Mahlein, A.-K. (2016). Plant disease detection by imaging sensors – Parallels and specific demands for precision agriculture and plant phenotyping. *Plant Disease, 100*(2), 241–251. https://doi.org/10.1094/PDIS-03-15-0340-FE

Mohanty, S. P., Hughes, D. P., & Salathé, M. (2016). Using deep learning for image-based plant disease detection. *Frontiers in Plant Science, 7*, 1419. https://doi.org/10.3389/fpls.2016.01419

Press Information Bureau. (2021, December 21). *Union Agriculture Minister releases Standard Operating Procedure (SOP) for use of drone in pesticide application for crop protection and for spraying soil and crop nutrients* [Press release]. Ministry of Agriculture & Farmers Welfare, Government of India. https://www.pib.gov.in/PressReleasePage.aspx?PRID=1783937

Press Information Bureau. (2023, April 20). *Crop specific SOPs issued for use of pesticides with farming drones* [Press release]. Ministry of Agriculture & Farmers Welfare, Government of India. https://www.pib.gov.in/PressReleasePage.aspx?PRID=1918341

Shagun. (2024, July 15). AI to the rescue: Cutting-edge tech to help cotton farmers fight pink bollworm. *Down To Earth*. https://www.downtoearth.org.in/agriculture/ai-to-the-rescue-cutting-edge-tech-to-help-cotton-farmers-fight-pink-bollworm

Stern, V. M., Smith, R. F., van den Bosch, R., & Hagen, K. S. (1959). The integration of chemical and biological control of the spotted alfalfa aphid: The integrated control concept. *Hilgardia, 29*(2), 81–101. https://doi.org/10.3733/hilg.v29n02p081

*All diagrams, examples and programs are original. Trap counts, nozzle values and the simulated field are illustrative; every calculation and the program output were checked by running them.*

## Practice MCQs

**Q1. Which optical technique measures leaf temperature to reveal stress?**

(a) RGB imaging (b) Thermography
(c) Chlorophyll fluorescence (d) LiDAR

**Q2. An AI output that draws a box and label around each insect on a trap image is**

(a) image classification (b) object detection
(c) semantic segmentation (d) clustering

**Q3. Labelling every pixel of a leaf image as healthy or lesion is**

(a) classification (b) detection
(c) semantic segmentation (d) regression

**Q4. If 9,000 of a leaf's 120,000 pixels are lesion pixels, disease severity is**

(a) 0.75% (b) 7.5%
(c) 13.3% (d) 75%

**Q5. Mohanty et al. (2016) found that their model's accuracy on images taken under different conditions fell to about**

(a) 99% (b) 85%
(c) 31% (d) 5%

**Q6. The economic threshold level is the pest population at which**

(a) the crop dies (b) control should begin to prevent economic injury
(c) pests disappear (d) yield doubles

**Q7. ICAR-CICR's pheromone-trap ETL for pink bollworm is**

(a) 2 moths per trap for 1 day (b) 8 moths per trap on 3 consecutive days
(c) 20 moths per hectare (d) 50 moths per week

**Q8. With daily trap counts 2, 3, 5, 9, 7, 8, 10, 12, the ETL of 8 moths for 3 consecutive days is first reached on day**

(a) 4 (b) 6
(c) 8 (d) 10

**Q9. Separating green plants from bare soil and residues is called**

(a) green-on-green detection (b) green-on-brown detection
(c) segmentation of roots (d) NDVI masking

**Q10. Patch spraying has shown herbicide savings of at least**

(a) 5% (b) 20%
(c) 50% (d) 95%

**Q11. A PWM nozzle with 0.8 L/min full flow, 0.5 m spacing and 6 km/h speed applies 160 L/ha; at 50% duty cycle it applies about**

(a) 40 L/ha (b) 80 L/ha
(c) 160 L/ha (d) 320 L/ha

**Q12. India's SOPs for using drones to apply pesticides and nutrients were first released in**

(a) 2014 (b) 2018
(c) December 2021 (d) 2025

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**Answers**

Q1 — (b) Thermography
Q2 — (b) object detection
Q3 — (c) semantic segmentation
Q4 — (b) 7.5%
Q5 — (c) 31%
Q6 — (b) control should begin to prevent economic injury
Q7 — (b) 8 moths per trap on 3 consecutive days
Q8 — (c) 8
Q9 — (b) green-on-brown detection
Q10 — (c) 50%
Q11 — (b) 80 L/ha
Q12 — (c) December 2021